

occurs at the instant when the wire is moving perpendicular to the field lines.

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Ans: C

Fig A, Mean Power, $W = \frac{V_{rms}^2}{R} = \frac{\left(\frac{V_0}{\sqrt{2}}\right)^2}{R} = \frac{V_0^2}{2R}$

Fig B, $V_{rms} = \sqrt{\frac{\text{area under the } V^2 - t \text{ curve over a period}}{T}}$

$$= \sqrt{\frac{(2V_0)^2 \frac{1}{2}T + 0}{T}} = \sqrt{2}V_0$$

Mean power, $P_B = \frac{V_{rms}^2}{R} = \frac{(\sqrt{2}V_0)^2}{R} = \frac{2V_0^2}{R} = 4W$

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Ans: C